2. A year 10 class investigated the reactions between some metals and hydrochloric acid. Their results are summarised in the table below.

Metal	Initial temperature (°C)	Final temperature (°C)	Rise in temperature (°C)	General observations
zinc	21	32	11	a few bubbles
calcium	22	66	44	lots of bubbles, solution spills out of test tube
magnesium	20		31	lots of bubbles
copper	21	21	0	no bubbles
iron	22	25	3	one or two bubbles

(a) (i) State which metal did **not** react with hydrochloric acid.

Give a reason for your choice. [1]

Metal ....

Reason ....

(ii) Calculate the final temperature for the reaction between magnesium and hydrochloric acid. [1]

Final temperature = ... °C

(b) What name is given to a reaction which gives a rise in temperature? [1]

Choose your answer from the box.

endothermic	**combustion**	**exothermic**	**precipitation**

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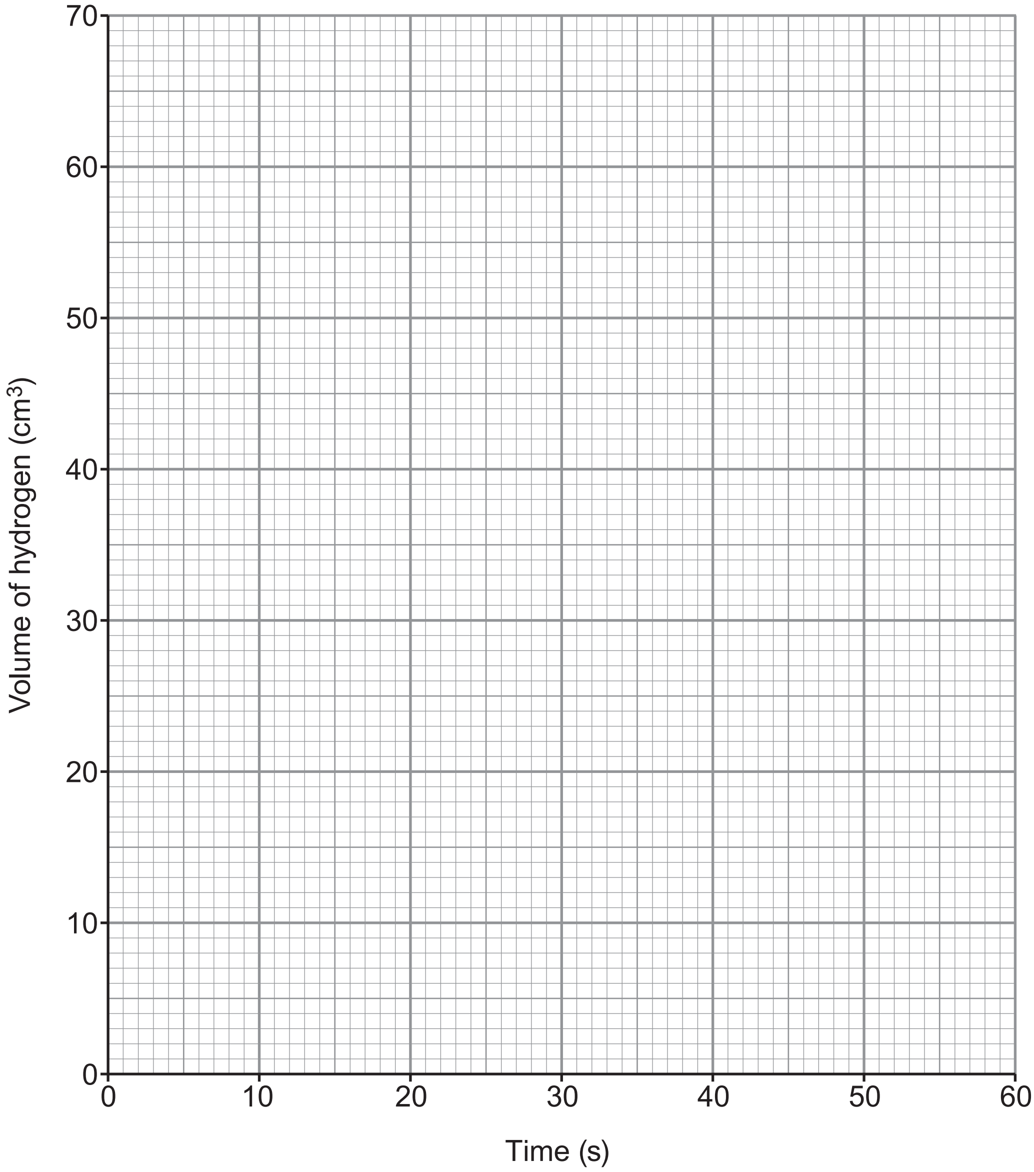
(c) The year 10 class then decided to investigate the rate of the reaction between magnesium and hydrochloric acid.

A piece of magnesium was placed in excess hydrochloric acid at 20 °C. The volume of hydrogen produced was recorded every 10 s.

The results obtained are shown in the table.

Time (s)	0	10	20	30	40	50	60
Volume of hydrogen (cm^3)	0	19	31	40	47	53	56

(i) Plot the volume of hydrogen against time on the grid. Draw a suitable line. [3]



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(ii) The reaction had not finished after 60 s. How does the graph show this?

Put a tick (✓) in the correct box. [1]

Graph stops at 60 s ☐

Graph is still rising at 60 s ☐

Graph reaches a maximum temperature of 56 °C ☐

(iii) Why does the reaction slow down over time?

Put a tick (✓) in the correct box. [1]

The particles collide with less energy so less chance of successful collisions ☐

The particles move slower so less chance of successful collisions ☐

The particles have less surface area so less chance of successful collisions ☐

The particles get used up so less chance of successful collisions ☐

(iv) Suggest **two** changes you could make to the hydrochloric acid to make the reaction faster. [2]

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(v) In the reaction between magnesium and hydrochloric acid, magnesium chloride is formed. Magnesium chloride contains Mg^{2+} ions and Cl^{-} ions.

Give the formula of magnesium chloride. [1]

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